cadence

Cadence Driver Verification Plan and Test Report

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Cadence Driver Verification Plan and Test Report

Cadence Design Systems

Hardware Identifier:cc9b6e1e2cc1d737b50b8e368e283aad

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Chapter 1. Overview

1.1. Document Purpose

The purpose of this document is to define the API testing for the Cadence driver firmware and to present the results.

1.2. Acronyms

API	Application Programming Interface.
Core Driver	Cadence Firmware component that provides IP programming abstraction.

Chapter 2. Test Equipment/Environment

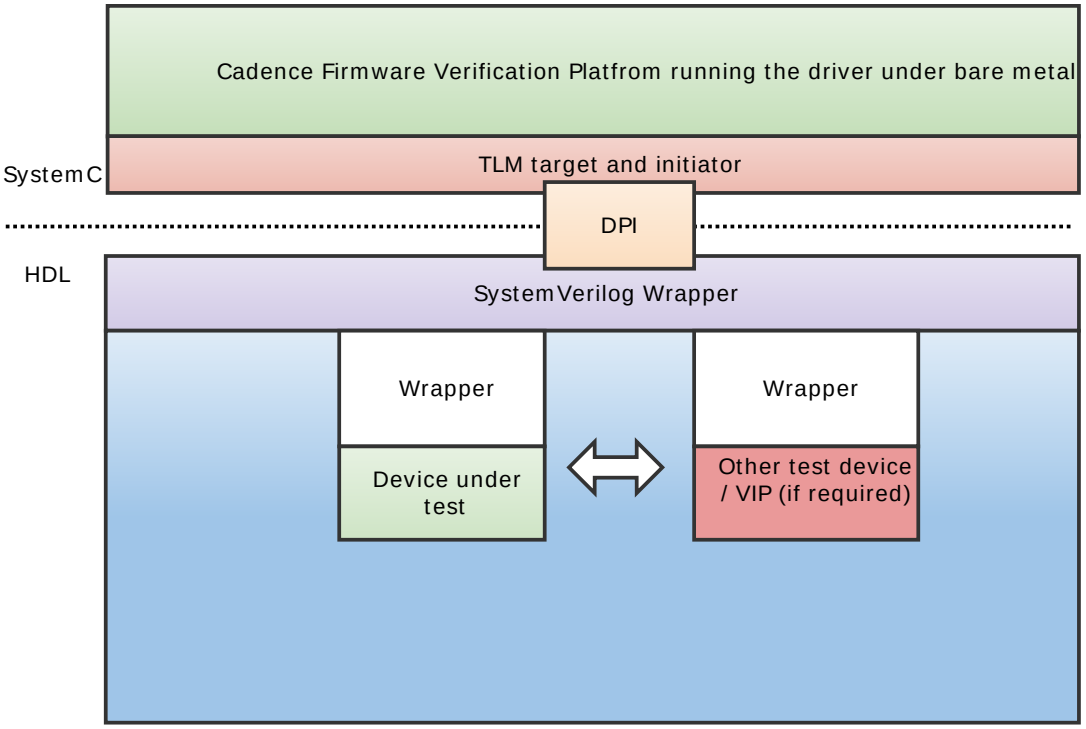
2.1. Overview

All testing is on a hardware simulated platform, running a bare-metal build. A sequence of tests is automatically generated as an executable program during the build process, and run from within the Cadence Firmware Verification Platform. Results are logged to a file via Cadence Firmware Verification Platform console ouptput then parsed and formatted into this report.

2.2. Cadence Firmware Verification Platform

The simulated hardware environment utilizes Cadence Firmware Verification Platform to connect the driver to the actual RTL for your HPNFC Controller.

Figure 2.1. Environment for verification of the HPNFC FW driver



2.3. Driver Tests

These tests confirm the operation of the core driver API under a range of different inputs, ensuring that operations comply with that described in the user documentation and that there are no unexpected errors. Some tests may not be appropriate for some APIs, depending on configuration, if so these will be marked as "NOT SUPPORTED". For each executed test, the result will be displayed as "PASSED" or "FAILED". Tests include:

Null pointer tests	API is checked to confirm that all null pointers are detected, and return appropriate error code.

Get/set and register access tests	API is checked for all get/set or register access functions, and confirm that these can correctly write and read the value.
Ranges tests	API is checked with valid and invalid parameters, and return code compared with that expected.
Functional tests	A sequence of predefined tests is run, with an expected result. These tests exercise the Cadence IP core, and may require use of additional test IP in the Cadence Firmware Verification Platform

NOTE: Not all configurations will require all types of test.

Chapter 3. Test Results

3.1. Test Summary

Functional test: 25 calls 24 passed 1 not supported

nullptr: 65 calls, 65 passed

getset: 18 calls, 16 passed, 2 not supported

ranges: 65 calls, 54 passed, 11 not supported

3.2. Test Details

3.2.1. Functional test

```
Multiplane test PASSED
Cache test PASSED
CDMA parse test PASSED
CDMA work mode test PASSED
CDMA work mode test 2 PASSED
CDMA chain descriptors test PASSED
CDMA sync feature test PASSED
Interleaved LUN test NOT SUPPORTED
ECC error injection test PASSED
ECC test PASSED
ECC scrambler test PASSED
PIO workmode test PASSED
Generic workmode test PASSED
Synchronous mode test PASSED
Extended status Test test PASSED
Debug interface test PASSED
Data integrity (CRC) test PASSED
Data integrity (parity of context memory) test PASSED
DMA trigger level test PASSED
Seed initial test PASSED
Remap configuration test PASSED
Remapping address test PASSED
Data integrity (parity of reamp memory) test PASSED
Timeout test PASSED
pSLC ONFI test PASSED
```

3.2.2. nullptr

```
probe PASSED
init PASSED
isr PASSED
start PASSED
stop PASSED
destroy PASSED
assembleCdmaDesc PASSED
execCmdReq PASSED
setWorkMode PASSED
setEccConfig PASSED
getEccConfig PASSED
setTransferConfig PASSED
getTransferConfig PASSED
cdmaGetFailedDesc PASSED
```

```
cdmaGetNextFailedDesc PASSED
cdmaDescGetOperationResult PASSED
reqGetOperationResult PASSED
cdmaParseDesc PASSED
setStatusPollingConfig PASSED
getStatusPollingConfig PASSED
setAutoStatusConfig PASSED
getAutoStatusConfig PASSED
getDeviceLayout PASSED
setDeviceSubLayout PASSED
setLunInterleavedConfig PASSED
getLunInterleavedConfig PASSED
getRnb PASSED
getBootStatus PASSED
getMultiplaneConfig PASSED
setMultiplaneConfig PASSED
getCacheConfig PASSED
setCacheConfig PASSED
getBchConfigInfo PASSED
getControllerInfo PASSED
getCdmaDescSize PASSED
getDebugData PASSED
setDebugIfType PASSED
getDebugIfType PASSED
setDiConfig PASSED
getDiConfig PASSED
setDiErrInjConfig PASSED
getDiErrInjConfig PASSED
setDmaConfig PASSED
getDmaConfig PASSED
writeRemapRecord PASSED
readRemapRecord PASSED
clearRemapTable PASSED
getRemapRecordCount PASSED
setRemapMask PASSED
getRemapMask PASSED
setRemapEnable PASSED
getRemapEnable PASSED
setPSlcConfig PASSED
getPSlcConfig PASSED
setZqCalConfig PASSED
getZqCalConfig PASSED
setZqCalEnable PASSED
getZqCalEnable PASSED
getZqStat PASSED
clearZqStat PASSED
setTimeoutConfig PASSED
getTimeoutConfig PASSED
setCePinReduction PASSED
getCePinReduction PASSED
getDeviceInfo PASSED
```

3.2.3. getset

```
EccConfig NOT SUPPORTED
TransferConfig PASSED
StatusPollingConfig PASSED
AutoStatusConfig PASSED
LunInterleavedConfig PASSED
MultiplaneConfig PASSED
CacheConfig PASSED
DebugIfType PASSED
DiConfig PASSED
```

```
DiErrInjConfig PASSED
DmaConfig PASSED
RemapMask PASSED
RemapEnable PASSED
PSlcConfig PASSED
ZqCalConfig PASSED
ZqCalEnable PASSED
TimeoutConfig PASSED
CePinReduction NOT SUPPORTED
```

3.2.4. ranges

```
probe NOT SUPPORTED
init NOT SUPPORTED
isr PASSED
start PASSED
stop PASSED
destroy PASSED
assembleCdmaDesc PASSED
execCmdReq NOT SUPPORTED
setWorkMode NOT SUPPORTED
setEccConfig NOT SUPPORTED
getEccConfig PASSED
setTransferConfig PASSED
getTransferConfig PASSED
cdmaGetFailedDesc PASSED
cdmaGetNextFailedDesc PASSED
cdmaDescGetOperationResult PASSED
reqGetOperationResult NOT SUPPORTED
cdmaParseDesc PASSED
setStatusPollingConfig PASSED
getStatusPollingConfig PASSED
setAutoStatusConfig PASSED
getAutoStatusConfig PASSED
getDeviceLayout PASSED
setDeviceSubLayout PASSED
setLunInterleavedConfig PASSED
getLunInterleavedConfig PASSED
getRnb PASSED
getBootStatus PASSED
getMultiplaneConfig PASSED
setMultiplaneConfig PASSED
getCacheConfig PASSED
setCacheConfig PASSED
getBchConfigInfo PASSED
getControllerInfo PASSED
getCdmaDescSize PASSED
getDebugData NOT SUPPORTED
setDebugIfType PASSED
getDebugIfType PASSED
setDiConfig PASSED
getDiConfig PASSED
setDiErrInjConfig PASSED
getDiErrInjConfig PASSED
setDmaConfig PASSED
getDmaConfig PASSED
writeRemapRecord NOT SUPPORTED
readRemapRecord NOT SUPPORTED
clearRemapTable NOT SUPPORTED
getRemapRecordCount PASSED
setRemapMask PASSED
getRemapMask PASSED
setRemapEnable PASSED
```

```
getRemapEnable PASSED
setPSlcConfig PASSED
getPSlcConfig PASSED
setZqCalConfig PASSED
getZqCalConfig PASSED
setZqCalEnable PASSED
getZqCalEnable PASSED
getZqStat PASSED
clearZqStat PASSED
setTimeoutConfig PASSED
getTimeoutConfig PASSED
setCePinReduction PASSED
getCePinReduction PASSED
getDeviceInfo NOT SUPPORTED
```

Chapter 4. Parasoft DTP Engine for C/C++ Analysis

```
Summary:
Parasoft DTP Engine for C/C++ Analysis: 53 rules detected
0 violations.

Checked rules:
All exit paths from a function with non-void return type shall have an
     explicit return statement with an expression
All member variables should be initialized in constructor
Always catch exceptions
Always provide empty brackets ([]) for delete when deallocating arrays
Avoid accessing arrays out of bounds
Avoid buffer overflow due to defining incorrect format limits
Avoid calling virtual functions from constructors
Avoid conditions that always evaluate to the same value
Avoid division by zero
Avoid double locking
Avoid function duplication
Avoid infinite loops
Avoid null pointer dereferencing
Avoid overflow due to reading a not zero terminated string
Avoid overflow when reading from a buffer
Avoid overflow when writing to a buffer
Avoid race conditions when using fork and file descriptors
Avoid race conditions while checking for the existence of a symbolic link
Avoid use before initialization
Boolean condition always evaluates to the same value due to enumeration with
     only zero or only non-zero constants
Class cannot inherit other class more than once unless it is virtual
     inheritance
Declare a copy assignment operator for classes with dynamically allocated
     memory
Declare a copy constructor for classes with dynamically allocated memory
Do not abandon unreleased locks
Do not acquire locks in different order
Do not call delete on non-pointers
Do not cast from or to incomplete class at the point of casting
Do not check for null after dereferencing
Do not delete objects with incomplete class at the point of deletion
Do not free resources using invalid pointers
Do not invoke malloc/realloc for objects having constructors
Do not pass negative values to functions expecting non-negative arguments
Do not throw from within destructor
Do not use 'delete' on pointers to a void type
Do not use blocking functions while holding a lock
Do not use resources that have been freed
Do not use string literals as operands of equality or relational operators
Ensure resources are freed
Floating-point expressions shall not be tested for equality or inequality
If a class has virtual functions it shall have a virtual destructor
Local variables should not use the same names as member variables
McCabe Cyclomatic Complexity
Nested Blocks Depth
Never provide brackets ([]) for delete when deallocating non-arrays
Pass objects by reference instead of by value
Pointer arithmetic performed on freshly allocated memory
Pointer should not be compared with NULL using relational operators
     &lt;, >, >=, &lt;=
Properly terminate character strings
Restore stream format
Suspicious argument to malloc
Suspicious setting of stream flags
```

```
The address of an object with automatic storage shall not be returned from a
     function
Throw by value, catch by reference
```